Enterprise Database Administration: Tablespace Management Assignment

Task 1: Creating and Configuring Permanent Tablespaces

The development team is introducing a new analytics module. To prevent analytical queries from locking or slowing down operational transactions, you need to isolate this data.

1. **Create a new permanent tablespace named ANALYTICS_DATA_TS.**
2. **Configure it with an initial datafile size of 500MB.**
3. **Enable auto-extending with a maximum limit of 5GB and an increment size of 100MB.**

ANSWER: -

SQL> CREATE TABLESPACE ANALYTICS_DATA_TS DATAFILE '/u01/app/oracle/oradata/KOL/adts01.dbf' SIZE 500M AUTOEXTEND ON NEXT 100M MAXSIZE 5G;

Tablespace created.

Check tablespace status

```
SQL> @/u03/tablespace_status.sql

   FILE_ID TABLESPACE_NAME                FILE_NAME                                                  MAX_SIZE(GB) CURR_SIZE(MB) STATUS    ONLINE_
---------- ------------------------------ ---------------------------------------------------------- ------------ ------------- --------- -------
         1 SYSTEM                         /u01/app/oracle/oradata/KOL/system01.dbf                     31.9999847           910 AVAILABLE SYSTEM
         3 SYSAUX                         /u01/app/oracle/oradata/KOL/sysaux01.dbf                     31.9999847           580 AVAILABLE ONLINE
         4 UNDOTBS1                       /u01/app/oracle/oradata/KOL/undotbs01.dbf                    31.9999847           300 AVAILABLE ONLINE
         5 ANALYTICS_DATA_TS              /u01/app/oracle/oradata/KOL/adts01.dbf                               5           500 AVAILABLE ONLINE
         7 USERS                          /u01/app/oracle/oradata/KOL/users01.dbf                      31.9999847             5 AVAILABLE ONLINE
```

Task 2: Managing Datafiles and Storage Growth

During a load test, the ANALYTICS_DATA_TS tablespace fills up faster than anticipated due to bulk data imports.

1. **Check current usage: Write or outline the SQL query used to monitor tablespace utilization and free space percentage.**
2. **Expand storage: Add a second datafile (analytics_data_02.dbf) of size 1GB to the ANALYTICS_DATA_TS tablespace.**
3. **Manual Resize: Manually resize the first datafile (analytics_data_01.dbf) to 2GB to accommodate immediate growth.**

ANSWER: -

POINT 1

```
SELECT tablespace_name,
    ROUND (used_space * 8192 / 1024 / 1024, 2)  AS used_mb,
    ROUND (tablespace_size * 8192 / 1024 / 1024, 2) AS max_size_mb,
    ROUND (used_percent, 2)                 AS pct_used,
```

ROUND (100 - used_percent, 2) AS pct_free

FROM dba_tablespace_usage_metrics WHERE TABLESPACE_NAME ='ANALYTICS_DATA_TS' ORDER BY used_percent DESC;

```
SQL> SELECT tablespace_name,
       ROUND(used_space * 8192 / 1024 / 1024, 2)  AS used_mb,
       ROUND(tablespace_size * 8192 / 1024 / 1024, 2) AS max_size_mb,
       ROUND(used_percent, 2)                     AS pct_used,
       ROUND(100 - used_percent, 2)               AS pct_free
FROM   dba_tablespace_usage_metrics
WHERE TABLESPACE_NAME ='ANALYTICS_DATA_TS'
ORDER  BY used_percent DESC;  2    3    4    5    6    7    8

TABLESPACE_NAME                USED_MB MAX_SIZE_MB   PCT_USED   PCT_FREE
------------------------------ ---------- ---------- ---------- ----------
ANALYTICS_DATA_TS                       1        5120        .02      99.98
```

POINT 2

SQL> ALTER TABLESPACE ANALYTICS_DATA_TS ADD DATAFILE '/u01/app/oracle/oradata/KOL/adts02.dbf' SIZE 1G AUTOEXTEND ON;

Tablespace altered.

```
SQL> @/u03/tablespace_status.sql

   FILE_ID TABLESPACE_NAME                FILE_NAME                                                      MAX_SIZE(GB) CURR_SIZE(MB) STATUS    ONLINE_
---------- ------------------------------ -------------------------------------------------------------- ------------ ------------- --------- -------
         1 SYSTEM                         /u01/app/oracle/oradata/KOL/system01.dbf                         31.9999847           910 AVAILABLE SYSTEM
         2 ANALYTICS_DATA_TS              /u01/app/oracle/oradata/KOL/adts02.dbf                           31.9999847          1024 AVAILABLE ONLINE
         3 SYSAUX                         /u01/app/oracle/oradata/KOL/sysaux01.dbf                         31.9999847           580 AVAILABLE ONLINE
         4 UNDOTBS1                       /u01/app/oracle/oradata/KOL/undotbs01.dbf                        31.9999847           300 AVAILABLE ONLINE
         5 ANALYTICS_DATA_TS              /u01/app/oracle/oradata/KOL/adts01.dbf                                 5           500 AVAILABLE ONLINE
         7 USERS                          /u01/app/oracle/oradata/KOL/users01.dbf                          31.9999847             5 AVAILABLE ONLINE
```

POINT 3

SQL> ALTER DATABASE DATAFILE 5 RESIZE 2G;

Database altered.

```
SQL> @/u03/tablespace_status.sql

   FILE_ID TABLESPACE_NAME                FILE_NAME                                                      MAX_SIZE(GB) CURR_SIZE(MB) STATUS    ONLINE_
---------- ------------------------------ -------------------------------------------------------------- ------------ ------------- --------- -------
         1 SYSTEM                         /u01/app/oracle/oradata/KOL/system01.dbf                         31.9999847           910 AVAILABLE SYSTEM
         2 ANALYTICS_DATA_TS              /u01/app/oracle/oradata/KOL/adts02.dbf                           31.9999847          1024 AVAILABLE ONLINE
         3 SYSAUX                         /u01/app/oracle/oradata/KOL/sysaux01.dbf                         31.9999847           590 AVAILABLE ONLINE
         4 UNDOTBS1                       /u01/app/oracle/oradata/KOL/undotbs01.dbf                        31.9999847           300 AVAILABLE ONLINE
         5 ANALYTICS_DATA_TS              /u01/app/oracle/oradata/KOL/adts01.dbf                                 5          2048 AVAILABLE ONLINE
         7 USERS                          /u01/app/oracle/oradata/KOL/users01.dbf                          31.9999847             5 AVAILABLE ONLINE

6 rows selected.
```

Task 3: Space Troubleshooting & Object Migration

A legacy table named CUSTOMER_LOGS located in the USERS tablespace is consuming excessive space and causing I/O bottlenecks.

1. **Create a dedicated high-performance tablespace named ARCHIVE_TS.**
2. **Move the CUSTOMER_LOGS table and its associated indexes from the USERS tablespace into ARCHIVE_TS with minimal downtime.**
3. **Rebuild or verify the validity of the indexes after the migration.**

ANSWER: -

USERNAME => DEV_APP_USER

TABLE_NAME => CUSTOMRES_LOG

INDEX_NAME = IDX_CUST_LOGS_CUSTID ||| IDX_CUST_LOGS_CREATED.

CREATE INDEX idx_cust_logs_custid ON customer_logs (customer_id);

CREATE INDEX idx_cust_logs_created ON customer_logs (created_at);

PART 1: -

CREATE TABLESPACE archive_ts DATAFILE '/u01/app/oracle/oradata/KOL/archive_ts01.dbf' SIZE 10G AUTOEXTEND ON NEXT 1G MAXSIZE 10G EXTENT MANAGEMENT LOCAL AUTOALLOCATE SEGMENT SPACE MANAGEMENT AUTO LOGGING DEFAULT ROW STORE COMPRESS ADVANCED;

```
SQL> @/u03/tablespace_status.sql

   FILE_ID TABLESPACE_NAME                FILE_NAME                                                    MAX_SIZE(GB) CURR_SIZE(MB) STATUS    ONLINE_
---------- ------------------------------ ------------------------------------------------------------ ------------ ------------- --------- -------
         1 SYSTEM                         /u01/app/oracle/oradata/KOL/system01.dbf                       31.9999847           910 AVAILABLE SYSTEM
         3 SYSAUX                         /u01/app/oracle/oradata/KOL/sysaux01.dbf                       31.9999847           600 AVAILABLE ONLINE
         4 UNDOTBS1                       /u01/app/oracle/oradata/KOL/undotbs01.dbf                      31.9999847           300 AVAILABLE ONLINE
         7 USERS                          /u01/app/oracle/oradata/KOL/users01.dbf                        31.9999847             5 AVAILABLE ONLINE
         8 ARCHIVE_TS                     /u01/app/oracle/oradata/KOL/archive_ts01.dbf                           10         10240 AVAILABLE ONLINE
```

PART 2: -

SELECT table_name, tablespace_name FROM user_tables WHERE table_name = 'CUSTOMER_LOGS';

```
SQL> SELECT table_name, tablespace_name FROM   user_tables WHERE  table_name = 'CUSTOMER_LOGS';

TABLE_NAME                     TABLESPACE_NAME
------------------------------ ------------------------------
CUSTOMER_LOGS                  USERS
```

SELECT table_name, tablespace_name FROM user_tables WHERE table_name = 'CUSTOMER_LOGS';

Online move (12.2+), the minimal-downtime route

From 12.2 onward, MOVE ONLINE keeps DML running during the move and maintains indexes automatically.

SQL> ALTER TABLE customer_logs MOVE ONLINE TABLESPACE archive_ts UPDATE INDEXES;

Table altered.

PART 3: -

SQL> ALTER INDEX pk_customer_logs REBUILD ONLINE TABLESPACE archive_ts;

Index altered.

SQL> ALTER INDEX idx_cust_logs_custid REBUILD ONLINE TABLESPACE archive_ts;

Index altered.

SQL> ALTER INDEX idx_cust_logs_created REBUILD ONLINE TABLESPACE archive_ts;

Index altered.

Now, validate the index & table information.

SQL> SELECT table_name, tablespace_name FROM user_tables WHERE table_name = 'CUSTOMER_LOGS';

SQL> SELECT index_name, tablespace_name, status FROM user_indexes WHERE table_name = 'CUSTOMER_LOGS';

```
SQL> SELECT table_name, tablespace_name FROM user_tables WHERE table_name = 'CUSTOMER_LOGS';

TABLE_NAME                     TABLESPACE_NAME
------------------------------ ------------------------------
CUSTOMER_LOGS                  ARCHIVE_TS

SQL>  SELECT index_name, tablespace_name, status FROM user_indexes WHERE table_name = 'CUSTOMER_LOGS';

INDEX_NAME                     TABLESPACE_NAME                STATUS
------------------------------ ------------------------------ --------
PK_CUSTOMER_LOGS               ARCHIVE_TS                     VALID
IDX_CUST_LOGS_CUSTID           ARCHIVE_TS                     VALID
IDX_CUST_LOGS_CREATED          ARCHIVE_TS                     VALID
```